

# Day3: Datatypes & Type-Conversions

What is Datatype?

## 1.Numaric Datatype

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→ Float and integer is called as numaric datatype

### 1.float

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→ A number which contains decimal values, we call it as a float datatype

Eg:

price=56.86

### 2.integer(int)

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→A normal number without any decimal values

Eg:

num=89

num\_2=7

## 2.String

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→String is a squnce of char that are enclosed in( " , ' , "" , '' ")

→ String is immutable

Eg:

any\_='Python is a language'

all\_='Ab,..\$#[]-+'

## 3.List

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→ List is a collection of different datatypes and it is represented by square brackets [ ]. That are separated by , inside the list we call it as items List is mutable.

Eg:

```
any_ = [1,'python',[5,6]]  
print(type(any_))
```

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#### 4.Tuple

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→ Tuple is a collection of different datatypes that are enclosed in Parentheses ( ) and those are separated by , and tuple is immutable

Eg:

```
nums =(1, 85.68,' python', [3,5] , (5,6))  
print(nums)
```

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#### 5.Dictionary

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→ Dictionary is collection of key : value pairs , keys and values are separated by colon ( : ). Key and value pair is called as item. and these items are separated by ( , ) Dictionary is represented using curly braces { }. In key place we can use immutable datatypes in values place we can use any datatype

Eg:

```
data_ = {1:2,  
         'name' : 'Praveen',  
         (2,3): 'tuple'}  
print(data_)
```

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#### 6.Set

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→ Set is a collection of unique elements and set can't allow any duplicate values inside it...

→ Set is represented by curly braces { } and the elements separated by ,

Eg:

```
an ={1,2,3}  
print(an)
```

## What is Type Conversion?

converting one data type into another data type

float → int ,Str

eg → int()

price= 45.68

print(int(prince))

→str()

price=45.78

con=str(price)

print(type(con))

int → float,str

eg: float()

num =90

print(float(num))

→ str

num =78

print(str(num))

print(type(num))

string → int, float

eg: → int()

do ='3456'

print(int(do))

→float

do ='59.63'

print(float(do))

list — tuple ,string

→ tuple()

```
nums =[1,2,3]  
print(tuple(nums))
```

tuple → list

eg → list

```
all_=(5,7,9)  
print(list(all_))
```

set → tuple, list

eg → tuple()

```
all_={5,6,7}  
print(tuple(all_))
```

dictionary → list

eg→dict()

```
details =[('name' , 'praveen') ,('edu' , 'B.sc')]  
print(dict(details))
```